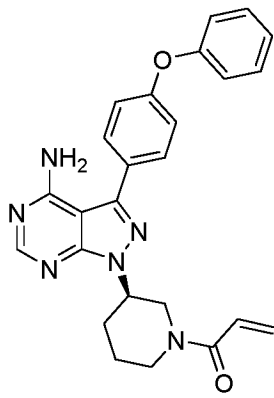


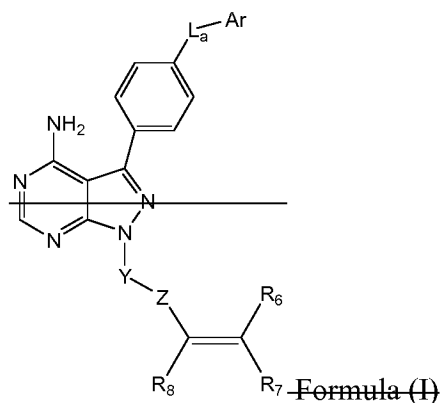
Amended Claims

1. A compound of the structure:



~~A method for use in~~ treating a disorder characterized by the presence or development of one or more solid tumors comprising administering to an individual in need thereof a ~~composition containing~~ a therapeutically effective amount of ~~the~~ a compound ~~that forms a covalent bond with a cysteine sidechain of a Bruton's tyrosine kinase or a Bruton's tyrosine kinase homolog.~~

2. ~~The method of claim 1, wherein the compound is a compound of Formula (I) having the structure:~~



wherein:

L_a is CH_2 , O, NH or S;

Ar is a substituted or unsubstituted aryl, or a substituted or unsubstituted heteroaryl; and either

(i) —Y is an optionally substituted group selected from among alkylene, heteroalkylene, arylene, heteroarylene, alkylenearylene, alkyleneheteroarylene, alkyleneecycloalkylene and alkyleneheterocycloalkylene;

Z is C(=O), NHC(=O), NR^aC(=O), NR^aS(=O)_x, where x is 1 or 2, and R^a is H, substituted or unsubstituted alkyl, or substituted or unsubstituted cycloalkyl; and

(a) —R₇ and R₈ are H;

R₆ is H, substituted or unsubstituted C₁-C₄alkyl, substituted or unsubstituted C₁-C₄heteroalkyl, C₁-C₈alkylaminoalkyl, C₁-C₈hydroxyalkylaminoalkyl, C₁-C₈alkoxyalkylaminoalkyl, substituted or unsubstituted C₃-C₆cycloalkyl, substituted or unsubstituted C₁-C₈alkylC₃-C₆cycloalkyl, substituted or unsubstituted aryl, substituted or unsubstituted C₂-C₈heterocycloalkyl, substituted or unsubstituted heteroaryl, C₁-C₄alkyl(aryl), C₁-C₄alkyl(heteroaryl), C₁-C₈alkylethers, C₁-C₈alkylamides, or C₁-C₄alkyl(C₂-C₈heterocycloalkyl); or

(b) —R₆ and R₈ are H;

R₇ is H, substituted or unsubstituted C₁-C₄alkyl, substituted or unsubstituted C₁-C₄heteroalkyl, C₁-C₈alkylaminoalkyl, C₁-C₈hydroxyalkylaminoalkyl, C₁-C₈alkoxyalkylaminoalkyl, substituted or unsubstituted C₃-C₆cycloalkyl, substituted or unsubstituted C₁-C₈alkylC₃-C₆cycloalkyl, substituted or unsubstituted aryl, substituted or unsubstituted C₂-C₈heterocycloalkyl, substituted or unsubstituted heteroaryl, C₁-C₄alkyl(aryl), C₁-C₄alkyl(heteroaryl), C₁-C₈alkylethers, C₁-C₈alkylamides, or C₁-C₄alkyl(C₂-C₈heterocycloalkyl); or

(c) —R₇ and R₈ taken together form a bond;

R₆ is H, substituted or unsubstituted C₁-C₄alkyl, substituted or unsubstituted C₁-C₄heteroalkyl, C₁-C₈alkylaminoalkyl, C₁-C₈hydroxyalkylaminoalkyl, C₁-C₈alkoxyalkylaminoalkyl, substituted or unsubstituted C₃-C₆cycloalkyl, substituted or unsubstituted C₁-C₈alkylC₃-C₆cycloalkyl, substituted or unsubstituted aryl, substituted or unsubstituted C₂-C₈heterocycloalkyl, substituted or unsubstituted heteroaryl, C₁-C₄alkyl(aryl), C₁-C₄alkyl(heteroaryl), C₁-C₈alkylethers, C₁-C₈alkylamides, or C₁-C₄alkyl(C₂-C₈heterocycloalkyl); or

(ii) Y is an optionally substituted group selected from cycloalkylene or heterocycloalkylene;

Z is C(=O), NHC(=O), NR^aC(=O), NR^aS(=O)_x, where x is 1 or 2, and R^a is H, substituted or unsubstituted alkyl, substituted or unsubstituted cycloalkyl; and either

(a) — R₇ and R₈ are H;

R₆ is substituted or unsubstituted C₁-C₄heteroalkyl, C₁-C₈hydroxyalkylaminoalkyl, C₁-C₈alkoxyalkylaminoalkyl, substituted or unsubstituted C₃-C₆cycloalkyl, substituted or unsubstituted C₁-C₈alkylC₃-C₆cycloalkyl, substituted or unsubstituted aryl, substituted or unsubstituted C₂-C₈heterocycloalkyl, substituted or unsubstituted heteroaryl, C₁-C₄alkyl(aryl), C₁-C₄alkyl(heteroaryl), C₁-C₈alkylethers, C₁-C₈alkylamides, or C₁-C₄alkyl(C₂-C₈heterocycloalkyl);

(b) — R₆ and R₈ are H;

R₇ is substituted or unsubstituted C₁-C₄heteroalkyl, C₁-C₈hydroxyalkylaminoalkyl, C₁-C₈alkoxyalkylaminoalkyl, substituted or unsubstituted C₃-C₆cycloalkyl, substituted or unsubstituted C₁-C₈alkylC₃-C₆cycloalkyl, substituted or unsubstituted aryl, substituted or unsubstituted C₂-C₈heterocycloalkyl, substituted or unsubstituted heteroaryl, C₁-C₄alkyl(aryl), C₁-C₄alkyl(heteroaryl), C₁-C₈alkylethers, C₁-C₈alkylamides, or C₁-C₄alkyl(C₂-C₈heterocycloalkyl); or

(c) — R₇ and R₈ taken together form a bond;

R₆ is substituted or unsubstituted C₁-C₄alkyl, substituted or unsubstituted C₁-C₄heteroalkyl, C₁-C₈alkylaminoalkyl, C₁-C₈hydroxyalkylaminoalkyl, C₁-C₈alkoxyalkylaminoalkyl, substituted or unsubstituted C₃-C₆cycloalkyl, substituted or unsubstituted C₁-C₈alkylC₃-C₆cycloalkyl, substituted or unsubstituted aryl, substituted or unsubstituted C₂-C₈heterocycloalkyl, substituted or unsubstituted heteroaryl, C₁-C₄alkyl(aryl), C₁-C₄alkyl(heteroaryl), C₁-C₈alkylethers, C₁-C₈alkylamides, or C₁-C₄alkyl(C₂-C₈heterocycloalkyl); or a

pharmaceutically active metabolite, pharmaceutically acceptable solvate, pharmaceutically acceptable salt, or a pharmaceutically acceptable prodrug thereof.

3. — The method of Claim 2, wherein:

L_a is O;

Ar is phenyl;

Z is C(=O), NHC(=O), or NCH₃C(=O).

4. ~~The method of claim 3, wherein~~

~~Y is alkyleneheterocycloalkylene;~~

~~R₆ and R₈ are H; and~~

~~R₇ is H, substituted or unsubstituted C₁-C₄alkyl, substituted or unsubstituted C₄-C₄heteroalkyl, C₄-C₈alkylaminoalkyl, C₄-C₈hydroxyalkylaminoalkyl, C₄-C₈alkoxyalkylaminoalkyl, substituted or unsubstituted C₃-C₆cycloalkyl, substituted or unsubstituted C₁-C₈alkylC₃-C₆cycloalkyl, substituted or unsubstituted aryl, substituted or unsubstituted C₂-C₈heterocycloalkyl, substituted or unsubstituted heteroaryl, C₁-C₄alkyl(aryl), C₁-C₄alkyl(heteroaryl), C₄-C₈alkylether, C₄-C₈alkylamide, or C₁-C₄alkyl(C₂-C₈heterocycloalkyl).~~

5. ~~The method of claim 3, wherein:~~

~~Y is an optionally substituted group selected from cycloalkylene or heterocycloalkylene; and~~

~~R₆ and R₈ are H; and~~

~~R₇ is substituted or unsubstituted C₁-C₄heteroalkyl, C₄-C₈hydroxyalkylaminoalkyl, C₄-C₈alkoxyalkylaminoalkyl, substituted or unsubstituted C₃-C₆cycloalkyl, substituted or unsubstituted C₁-C₈alkylC₃-C₆cycloalkyl, substituted or unsubstituted aryl, substituted or unsubstituted C₂-C₈heterocycloalkyl, substituted or unsubstituted heteroaryl, C₁-C₄alkyl(aryl), C₁-C₄alkyl(heteroaryl), C₄-C₈alkylether, C₄-C₈alkylamide, or C₁-C₄alkyl(C₂-C₈heterocycloalkyl).~~

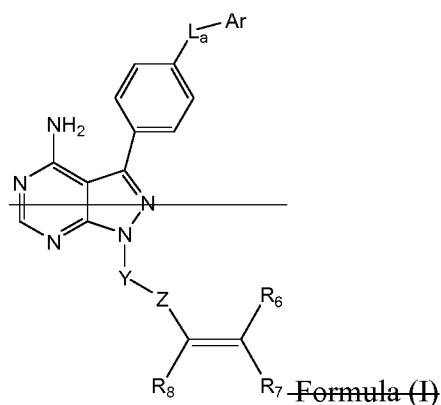
62. ~~The use method~~ compound for use according ~~of~~ claim 1, wherein the disorder is a sarcoma, lymphoma, or carcinoma.

37. The ~~use~~compound for use according to ~~method~~ of claim 1, wherein the disorder is ~~mammary ductal carcinoma, lobular carcinoma~~, an adenocarcinoma, small cell lung carcinoma, non-small cell lung carcinoma, a melanoma, or a combination thereof.

48. The ~~use~~~~method~~compound for use according to ~~of~~ claim 1, wherein the disorder is pancreatic cancer.

9. — A method for treating a disorder characterized by the presence or development of one or more solid tumors comprising administering to an individual in need thereof a composition containing a therapeutically effective amount of a compound that forms a covalent bond with a cysteine sidechain of a HER4.

10. — The method of claim 9, wherein the compound is a compound of Formula (I) having the structure:



wherein:

L_a is CH_2 , O, NH or S;

Ar is a substituted or unsubstituted aryl, or a substituted or unsubstituted heteroaryl; and either

(i) — Y is an optionally substituted group selected from among alkylene, heteroalkylene, arylene, heteroarylene, alkylenearylene, alkyleneheteroarylene, alkylenealkylene and alkyleneheteroalkylene;

Z is C(=O), NHC(=O), NR^aC(=O), NR^aS(=O)_x, where x is 1 or 2, and R^a is H, substituted or unsubstituted alkyl, substituted or unsubstituted cycloalkyl; and either

(a) ——— R₇ and R₈ are H;

R₆ is H, substituted or unsubstituted C₁-C₄alkyl, substituted or unsubstituted C₁-C₄heteroalkyl, C₁-C₈alkylaminoalkyl, C₁-C₈hydroxyalkylaminoalkyl, C₁-C₈alkoxyalkylaminoalkyl, substituted or unsubstituted C₃-C₆cycloalkyl, substituted or unsubstituted C₁-C₈alkylC₃-C₆cycloalkyl, substituted or unsubstituted aryl, substituted or unsubstituted C₂-C₈heterocycloalkyl, substituted or unsubstituted heteroaryl, C₁-C₄alkyl(aryl), C₁-C₄alkyl(heteroaryl), C₁-C₈alkylether, C₁-C₈alkylamide, or C₁-C₄alkyl(C₂-C₈heterocycloalkyl);

(b) ——— R₆ and R₈ are H;

R₇ is H, substituted or unsubstituted C₁-C₄alkyl, substituted or unsubstituted C₁-C₄heteroalkyl, C₁-C₈alkylaminoalkyl, C₁-C₈hydroxyalkylaminoalkyl, C₁-C₈alkoxyalkylaminoalkyl, substituted or unsubstituted C₃-C₆cycloalkyl, substituted or unsubstituted C₁-C₈alkylC₃-C₆cycloalkyl, substituted or unsubstituted aryl, substituted or unsubstituted C₂-C₈heterocycloalkyl, substituted or unsubstituted heteroaryl, C₁-C₄alkyl(aryl), C₁-C₄alkyl(heteroaryl), C₁-C₈alkylether, C₁-C₈alkylamide, or C₁-C₄alkyl(C₂-C₈heterocycloalkyl); or

(c) ——— R₇ and R₈ taken together form a bond;

R₆ is H, substituted or unsubstituted C₁-C₄alkyl, substituted or unsubstituted C₁-C₄heteroalkyl, C₁-C₈alkylaminoalkyl, C₁-C₈hydroxyalkylaminoalkyl, C₁-C₈alkoxyalkylaminoalkyl, substituted or unsubstituted C₃-C₆cycloalkyl, substituted or unsubstituted C₁-C₈alkylC₃-C₆cycloalkyl, substituted or unsubstituted aryl, substituted or unsubstituted C₂-C₈heterocycloalkyl, substituted

or unsubstituted heteroaryl, C₁-C₄alkyl(aryl), C₁-C₄alkyl(heteroaryl), C₁-C₈alkylether, C₁-C₈alkylamide, or C₁-C₄alkyl(C₂-C₈heterocycloalkyl); or

(ii) Y is an optionally substituted group selected from cycloalkylene or heterocycloalkylene;

Z is C(=O), NHC(=O), NR^aC(=O), NR^aS(=O)_x, where x is 1 or 2, and R^a is H, substituted or unsubstituted alkyl, substituted or unsubstituted cycloalkyl; and either

(a) ——— R₇ and R₈ are H;

R₆ is substituted or unsubstituted C₁-C₄heteroalkyl, C₁-C₈hydroxyalkylaminoalkyl, C₁-C₈alkoxyalkylaminoalkyl, substituted or unsubstituted C₃-C₆cycloalkyl, substituted or unsubstituted C₁-C₈alkylC₃-C₆cycloalkyl, substituted or unsubstituted aryl, substituted or unsubstituted C₂-C₈heterocycloalkyl, substituted or unsubstituted heteroaryl, C₁-C₄alkyl(aryl), C₁-C₄alkyl(heteroaryl), C₁-C₈alkylethers, C₁-C₈alkylamides, or C₁-C₄alkyl(C₂-C₈heterocycloalkyl);

(b) ——— R₆ and R₈ are H;

R₇ is substituted or unsubstituted C₁-C₄heteroalkyl, C₁-C₈hydroxyalkylaminoalkyl, C₁-C₈alkoxyalkylaminoalkyl, substituted or unsubstituted C₃-C₆cycloalkyl, substituted or unsubstituted C₁-C₈alkylC₃-C₆cycloalkyl, substituted or unsubstituted aryl, substituted or unsubstituted C₂-C₈heterocycloalkyl, substituted or unsubstituted heteroaryl, C₁-C₄alkyl(aryl), C₁-C₄alkyl(heteroaryl), C₁-C₈alkylethers, C₁-C₈alkylamides, or C₁-C₄alkyl(C₂-C₈heterocycloalkyl); or

(c) ——— R₇ and R₈ taken together form a bond;

R₆ is substituted or unsubstituted C₁-C₄alkyl, substituted or unsubstituted C₁-C₄heteroalkyl, C₁-C₈alkylaminoalkyl, C₁-C₈hydroxyalkylaminoalkyl, C₁-C₈alkoxyalkylaminoalkyl, substituted or

~~unsubstituted C₃-C₆cycloalkyl, substituted or unsubstituted C₄-C₈alkylC₃-C₆cycloalkyl, substituted or unsubstituted aryl, substituted or unsubstituted C₂-C₈heterocycloalkyl, substituted or unsubstituted heteroaryl, C₄-C₄alkyl(aryl), C₄-C₄alkyl(heteroaryl), C₄-C₈alkylethers, C₄-C₈alkylamides, or C₄-C₄alkyl(C₂-C₈heterocycloalkyl); and pharmaceutically active metabolites, or pharmaceutically acceptable solvates, pharmaceutically acceptable salts, or pharmaceutically acceptable prodrugs thereof.~~

11. ~~_____~~ The method of Claim 10, wherein:

~~L_a is O;~~

~~Ar is phenyl;~~

~~Z is C(=O), NHC(=O), or NCH₃C(=O).~~

12. ~~_____~~ The method of claim 11, wherein

~~Y is alkyleneheterocycloalkylene;~~

~~R₆ and R₈ are H; and~~

~~R₇ is H, substituted or unsubstituted C₄-C₄alkyl, substituted or unsubstituted C₄-C₄heteroalkyl, C₄-C₈alkylaminoalkyl, C₄-C₈hydroxyalkylaminoalkyl, C₄-C₈alkoxyalkylaminoalkyl, substituted or unsubstituted C₃-C₆cycloalkyl, substituted or unsubstituted C₄-C₈alkylC₃-C₆cycloalkyl, substituted or unsubstituted aryl, substituted or unsubstituted C₂-C₈heterocycloalkyl, substituted or unsubstituted heteroaryl, C₄-C₄alkyl(aryl), C₄-C₄alkyl(heteroaryl), C₄-C₈alkylether, C₄-C₈alkylamide, or C₄-C₄alkyl(C₂-C₈heterocycloalkyl).~~

13. ~~_____~~ The method of claim 11, wherein:

~~Y is an optionally substituted group selected from cycloalkylene or heterocycloalkylene; and~~

R₆ and R₈ are H; and

R₇ is substituted or unsubstituted C₁-C₄heteroalkyl, C₁-C₈hydroxyalkylaminoalkyl, C₁-C₈alkoxyalkylaminoalkyl, substituted or unsubstituted C₃-C₆cycloalkyl, substituted or unsubstituted C₁-C₈alkylC₃-C₆cycloalkyl, substituted or unsubstituted aryl, substituted or unsubstituted C₂-C₈heterocycloalkyl, substituted or unsubstituted heteroaryl, C₁-C₄alkyl(aryl), C₁-C₄alkyl(heteroaryl), C₁-C₈alkylethers, C₁-C₈alkylamides, or C₁-C₄alkyl(C₂-C₈heterocycloalkyl).

14. — The method of claim 9, wherein the disorder is mammary ductal carcinoma, lobular carcinoma, or a combination thereof.

15. — A method for treating a disorder characterized by the presence or development of one or more solid tumors comprising administering to an individual in need thereof a kinase inhibitor that selectively and irreversibly binds to a protein tyrosine kinase selected from Btk, a Btk homolog, an ACK, HER4, and a Btk kinase cysteine homolog, in which the kinase inhibitor reversibly and non-selectively binds to a multiplicity of protein tyrosine kinases, and further in which the plasma half life of the kinase inhibitor is less than about 4 hours.

16. — The method of claim 15, wherein the kinase inhibitor selectively and irreversibly binds to Btk.

17. — The method of claim 15, wherein the kinase inhibitor irreversibly binds to HER4.

18. — The method of claim 15, wherein the disorder is a sarcoma, lymphoma, and/or carcinoma.

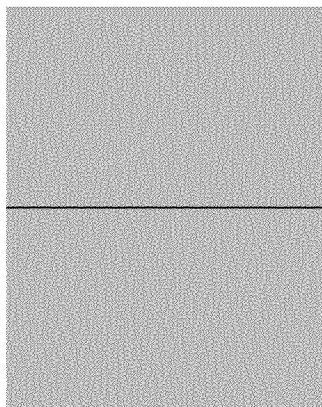
19. — The method of claim 15, wherein the disorder is mammary ductal carcinoma, lobular carcinoma, an adenocarcinoma, small cell lung carcinoma, non-small cell lung carcinoma, a melanoma, or a combination thereof.

20. — The method of claim 15, wherein the disorder is mammary ductal carcinoma, lobular carcinoma, or a combination thereof.

21. — The method of claim 15, wherein the disorder is pancreatic cancer.

22. — The method of claim 15, in which the plasma half life of the kinase inhibitor is less than about 3 hours.

23. — The method of claim 15, wherein the kinase inhibitor has the structure of Formula (VII):



Formula (VII)

wherein:

-  is a moiety that binds to the active site of a kinase, including a tyrosine kinase, further including a Btk kinase cysteine homolog;
- Y is an optionally substituted group selected from among alkylene, heteroalkylene, arylene, heteroarylene, heterocycloalkylene, cycloalkylene, alkylenearylene, alkyleneheteroarylene, alkyleneecycloalkylene, and alkyleneheterocycloalkylene;
- Z is C(=O), OC(=O), NHC(=O), NCH₂C(=O), C(=S), S(=O)_x, OS(=O)_x, NHS(=O)_x, where x is 1 or 2;

~~R₆ and R₈ are independently selected from among H, unsubstituted C₁-C₄alkyl, substituted C₁-C₄alkyl, unsubstituted C₁-C₄heteroalkyl, substituted C₁-C₄heteroalkyl, unsubstituted C₃-C₆cycloalkyl, substituted C₃-C₆cycloalkyl, unsubstituted C₂-C₆heterocycloalkyl, and substituted C₂-C₆heterocycloalkyl; or~~

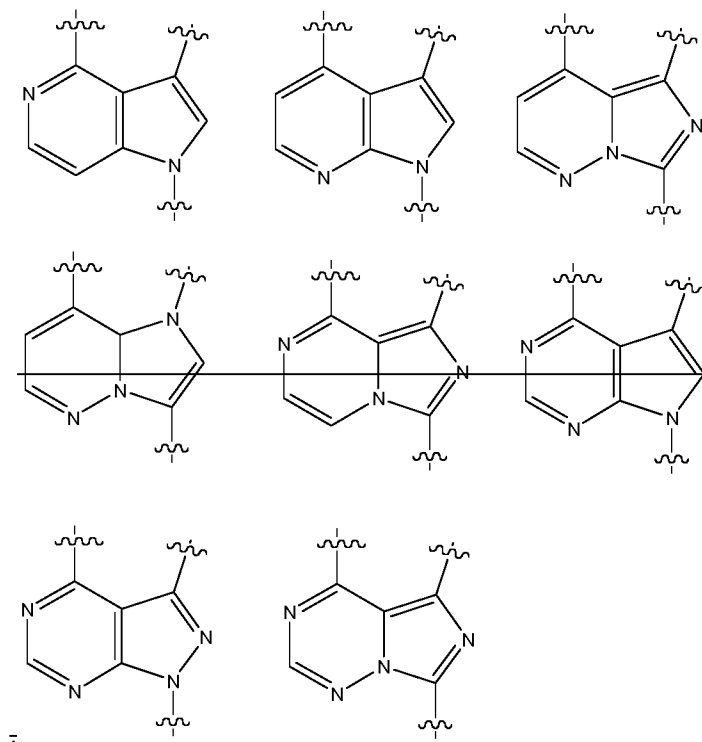
~~R₆ and R₈ taken together form a bond;~~

~~R₇ is H, substituted or unsubstituted C₁-C₄alkyl, substituted or unsubstituted C₁-C₄heteroalkyl, C₁-C₆alkoxyalkyl, C₁-C₈alkylaminoalkyl, C₁-C₈hydroxyalkylaminoalkyl, substituted or unsubstituted C₃-C₆cycloalkyl, substituted or unsubstituted aryl, substituted or unsubstituted C₂-C₈heterocycloalkyl, substituted or unsubstituted heteroaryl, C₁-C₄alkyl(aryl), C₁-C₄alkyl(heteroaryl), C₁-C₄alkyl(C₃-C₆cycloalkyl), or C₁-C₄alkyl(C₂-C₈heterocycloalkyl); or a~~

~~pharmaceutically active metabolite, or pharmaceutically acceptable solvate,~~

~~pharmaceutically acceptable salt, or pharmaceutically acceptable prodrug thereof.~~

24. — The method of claim 21, wherein  is a substituted fused biaryl moiety selected from



25. — The method of claim 22, wherein:

Z is C(=O), NHC(=O), NCH₃C(=O), or S(=O)₂;

~~each of R₆ and R₈ is H; or~~

~~R₆ and R₈ taken together form a bond.~~

26. ~~The method of claim 23, wherein:~~

~~Y is a 4-, 5-, 6-, or 7-membered cycloalkylene ring; or~~

~~Y is a 4-, 5-, 6-, or 7-membered heterocycloalkylene ring; or~~

~~Y is a C₁-C₄alkylene, or 4-, 5-, 6-, or 7-membered heterocycloalkylene ring;~~

~~R₇ is H, substituted or unsubstituted C₁-C₄alkyl, substituted or unsubstituted C₁-~~

~~C₄heteroalkyl, C₁-C₆alkoxyalkyl, C₁-C₈alkylaminoalkyl, C₁-~~

~~C₈hydroxyalkylaminoalkyl, C₁-C₈alkoxyalkylaminoalkyl, C₁-C₄alkyl(aryl), C₁-~~

~~C₄alkyl(heteroaryl), C₁-C₄alkyl(C₃-C₈cycloalkyl), or C₁-C₄alkyl(C₂-~~

~~C₈heterocycloalkyl).~~